

# MUHAMMAD HAMZA

AI Engineer | Generative AI, Agentic Systems & RAG

AI Engineer specializing in agentic AI systems, LLM application development, and Retrieval-Augmented Generation (RAG). Hands-on experience designing multi-agent workflows with LangChain and LangGraph, building production RAG pipelines over vector databases (FAISS, ChromaDB), and shipping LLM-powered automation for enterprise, cybersecurity, and backend systems. Background spans end-to-end AI development: data preprocessing, model training and evaluation, prompt engineering, API integration, and deployment, with peer-reviewed research in 3D brain MRI classification and medical image analysis, including skin lesion segmentation and diabetic retinopathy diagnosis. Experienced in owning AI projects across the complete delivery lifecycle, including direct client requirement discussions, solution architecture, hands-on technical team leadership, implementation, progress demonstrations, feedback cycles, deployment, and final handover.



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LOCATION  
Lahore, Pakistan

## EXPERIENCE

### AI Engineer

CyberZeus Software Systems | 08/2025 - Present  
Lahore, Pakistan

- Own end-to-end delivery of AI/ML solutions, from client requirement discovery and solution architecture through implementation, evaluation, deployment, and final handover.
- Conduct direct client meetings to understand business requirements, clarify technical and functional needs, define solution scope, demonstrate progress, gather feedback, and align delivery with client expectations.
- Lead a four-person development team of two junior developers and two interns, covering task decomposition, technical guidance, implementation reviews, integration support, troubleshooting, and delivery coordination.
- Architect and implement agentic AI and RAG systems, including LangGraph workflows, retrieval strategy, agent state, tool orchestration, context handling, and FAISS/ChromaDB knowledge retrieval.
- Translate client requirements into technical milestones while retaining ownership of architecture, integration quality, and final solution delivery across LLM, RAG, cybersecurity, backend automation, and intelligent workflow projects.

### AI Developer

Minhaj Solutions | 04/2024 - 08/2025  
Renala Khurd, Pakistan

- Built an agentic RAG-based support assistant that used LLMs to understand and resolve technical product queries.
- Replaced traditional rule-based responses with context-aware answers generated from relevant product knowledge.
- Developed the embedding, semantic search, and response-generation pipeline to provide real-time, knowledge-grounded support.
- Worked on applied AI and machine learning projects using real-world medical imaging and classification datasets.

### Freelance Python Developer & ML Engineer

Fiverr | 07/2023 - Present  
Remote

- Completed 12+ international projects in Python automation, machine learning, and data analysis as a Level 1 Fiverr Seller.
- Delivered end-to-end ML solutions including preprocessing, model training, evaluation, deployment, and reporting workflows.

## TECHNICAL SKILLS

### Agentic AI, Generative AI & LLM Orchestration

LangChain  
LangGraph (multi-agent workflows, state machines, tool use)  
Model Context Protocol (MCP)  
Large Language Models (LLMs)  
Prompt Engineering  
LLM Application Development

### Machine Learning & Deep Learning

Machine Learning (ML)  
Deep Learning (DL)  
Natural Language Processing (NLP)  
Model Training & Evaluation  
Data Preprocessing  
CNN, TabNet, Transformers

### RAG, Vector Search & Information Retrieval

RAG Pipeline Architecture  
LlamaIndex  
Haystack  
Vector Search & Databases (FAISS, ChromaDB)  
Embedding Generation  
Semantic Search  
Document Chunking & Indexing

### Programming, Cloud & DevOps

Python, C++  
NumPy, Pandas, Matplotlib, Seaborn  
Jupyter Notebook, VS Code, GitHub  
AWS  
GitHub Actions, CI/CD, Jenkins  
Docker, Kubernetes  
Postman

## EDUCATION

### BS Software Engineering

University of Management and Technology (UMT)  
2020 - 2024

### FSC

Punjab Group of Colleges  
2017 - 2019

PROJECTS

Agentic Cybersecurity AI Assistant — Multi-Agent RAG System

Enterprise-grade AI cybersecurity assistant built with LangChain and LangGraph, combining multi-agent orchestration, Retrieval-Augmented Generation (RAG), vector search, external security tools, and automated reporting to support vulnerability analysis and continuous monitoring.

- Architected multi-agent workflows with specialized planner, retriever, and analyst agents for automated vulnerability analysis, reasoning, and report generation.
- Engineered the RAG pipeline using FAISS/ChromaDB, embeddings, semantic search, and vulnerability knowledge bases to provide context-grounded security analysis.
- Designed fallback/error-handling strategies for failed tool calls and incomplete retrieval results.
- Optimized token utilization and inference latency through context-window management, selective retrieval, and efficient prompt construction.
- Implemented persistent session memory and contextual state management to maintain continuity across multi-step agent workflows.
- Implemented CI/CD pipelines using GitHub Actions and containerized agent services with Docker.

AI-Based Ticket Scalping Prevention System

- Developed an AI-based fraud detection and dynamic pricing system for ticketing platforms.
- Built anomaly detection models to identify bot activity and suspicious transactions.
- Implemented pricing and fraud prevention mechanisms for fair ticket distribution.
- Implemented CI/CD workflows via GitHub Actions to automate model retraining, testing, and deployment to production.

AI-Driven Intrusion Detection for Critical Systems

- Designed a hybrid deep-learning intrusion detection architecture combining TabNet and Transformers.
- Trained and evaluated models on the CIC-IDS 2017 cybersecurity dataset.
- Applied attention-based feature learning to improve classification of malicious network-traffic patterns.
- Deployed the detection model using Docker and Kubernetes for scalable, real-time monitoring across critical infrastructure nodes.

Agentic RAG-Powered Support Assistant

- Designed an agentic RAG-powered support assistant that autonomously resolves technical product queries using LLM-driven reasoning.
- Implemented retrieval-augmented, context-aware response generation in place of rule-based query handling.
- Built embedding, semantic search, and automated response-generation pipelines powering real-time, knowledge-grounded conversations.

CERTIFICATIONS & ACHIEVEMENTS

Fully Funded Scholar — Design & Development of Unmanned Aerial Systems

Samara National Research University, Russia | Nomination by the University of Okara | 2025

Kaggle Notebook Expert Badge, Kaggle

Ranked Top 2,100 of ~60,000+ contributors in Kaggle Notebooks.

RESEARCH & PUBLICATIONS

3D-MobiBrainNet: Multi-class Alzheimer's disease classification using 3D brain magnetic resonance imaging

*Ain Shams Engineering Journal*, Volume 16, Issue 11, 2025, Article 103714.  
Zia-ur Rehman, Mohd Khalid Awang, Ghulam Ali, **Muhammad Hamza**, Tariq Ali, Muhammad Ayaz, Mohammad Hijji

DOI: 10.1016/j.asej.2025.103714

A computationally efficient 3D deep learning framework for multi-class Alzheimer’s disease classification using volumetric brain MRI.

WIDENet: A Novel Lightweight CNN for Robust Skin Lesion Segmentation

*IEEE Access*, Volume 14, 2026, pp. 57763–57781.  
Ghulam Ali, Mohd Khalid Awang, Javed Rashid, **Muhammad Hamza**, Osama A. Khashan, Anwar Ghani

DOI: 10.1109/ACCESS.2026.3683571

A lightweight medical image segmentation framework combining multiresolution wavelets, transformer-based contextual encoding, and attentive feature refinement.

Hierarchical deep learning with stability-driven normalization and regularization for accurate multi-class diabetic retinopathy diagnosis

*Ain Shams Engineering Journal*, Volume 17, Issue 12, 2026, Article 104453.  
Ghulam Ali, Mohd Khalid Awang, Tariq Ali, Mohammad Hijji, Muhammad Ayaz, **Muhammad Hamza**, Azqa Jafar

DOI: 10.1016/j.asej.2026.104453

A compact hierarchical vision encoder combining convolutional features with stability-driven normalization and regularization for five-class diabetic retinopathy diagnosis on the APTOS dataset.